

A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file star5_small. [Note: The data file star5 contains more observations and variables.]

a. Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?

$$\beta_1 \approx 916.442$$

$$\beta_2 \approx 12.175$$

$$p\text{-value} \approx 0.02363$$

The intercept(916.442) represents the average total test score for students in regular-sized classes. The coefficient on SMALL(12.175) indicates that students in small classes score about 12.175 points higher on average than students in regular classes. Since the p-value(0.0236) is less than 0.05, the effect is statistically significant. Therefore, the results suggest that smaller class sizes improve student learning outcomes.

b. Repeat part (a) using dependent variables READSCORE and MATHSCORE. Do you observe any differences?

	TOTALSCORE
β_1	916.442
β_2	12.175
p-value	0.02363

	READSCORE
β_1	432.665
β_2	6.924
p-value	0.001507

	MATHSCORE
β_1	483.777
β_2	5.251
p-value	0.1427

The results show that students in small classes score about 6.9 points higher in reading, and this effect is statistically significant. For math scores, students in small classes score about 5.3 points higher, but the effect is not statistically significant. Thus, small class size appears to have a stronger effect on reading than on math.

c. Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?

$$\gamma_1 \approx 916.442$$

$$\gamma_2 \approx 4.306$$

$$p\text{-value} \approx 0.3888$$

The intercept(916.442) represents the average total score for students in regular-sized classes without a teacher aide. The coefficient on AIDE(4.306) indicates that students in classes with a teacher aide score about 4.3 points higher on average. However, the effect is not statistically significant, suggesting that adding a teacher aide does not have a meaningful impact on student learning outcomes.

Compared with the effect of small classes, the impact of adding a teacher aide appears to be much smaller and not statistically significant.

d. Repeat part (c) using dependent variables READSCORE and MATHSCORE. Do you observe any differences?

	TOTALSCORE
γ_1	916.442
γ_2	4.306
p-value	0.3888

	READSCORE
γ_1	432.665
γ_2	2.871
p-value	0.1615

	MATHSCORE
γ_1	483.777
γ_2	1.435
p-value	0.6642

The p-values are greater than 0.05, indicating that these effects are not statistically significant. Similar to the result for TOTALSCORE, the presence of a teacher aide does not appear to have a significant impact on student learning.